

Spectral semantics of distributed systems

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This document relies on concepts and results discussed in my Research Statement (RS)

The goal of my project is to develop a uniform approach to the question of *distributed computability*, that is, the problem of characterising and comparing which tasks can be solved by which distributed protocols. In sequential computing, a wide variety of abstract and universal approaches to questions of computability have been developed, *e.g.* via Turing machines, the Lambda calculus, or semantic approaches such as domain theory. When it comes to distributed computing, while many theoretical advances and insights have been made over the past thirty to forty years, a general framework is lacking. One may distinguish two main currents for the formal treatment of such systems. On the one hand, event structures, higher dimensional automata, and Petri nets provide a means of encoding the *scheduling constraints* of the protocol, *i.e.* the possible temporal scheduling order of concurrent executions. On the other, determining obstructions to task-solvability by tracking the *epistemic dynamics*, *i.e.* the evolution of knowledge in the system, has been approached through logical, combinatorial and topological methods. Although scheduling and epistemics are inherently linked in distributed systems, their relationship has of yet not been systematically studied.

Different scheduling constraints, but also other modalities, such as how much and which type of information is communicated, result in a plethora of models of distributed protocols and tasks. Due to the lack of a systematic method translating these constraints into a formal object characterising their distributed computability power, the description and comparison thereof has thus far been handled in a piecemeal context-based manner. A recent approach to distributed computability describes task-solvability using topological objects called simplicial complexes, winning Herlihy and Shavit the Gödel prize in 2004. However, their groundbreaking computability result, the Asynchronous Computability Theorem (ACT) [26], applies only to a restricted model of communication, namely Iterated Immediate Snapshot (IIS). Translating its informational and round-based scheduling constraints into transformations of an epistemic space, they characterise its task-solvability, and then use ad-hoc algorithmic means to extend the result to a more general asynchronous model. This result is thus also specific to one model of communication, and a context-based reduction.

In [9], we established a surprising new connection between the simplicial approach of Herlihy and Shavit and non-Hausdorff topologies, obtaining a generalisation of the ACT beyond IIS, characterising computability for a wider class of round-based protocols via spectral topologies and Stone duality. The goal of my project is to use this result as the basis for a *general topological framework* addressing questions of distributed computability. I propose a systematic means of *describing* and *comparing* the distributed computability of a large class of models. This will be achieved by extending spectral semantics to a number of settings (Section 1) and then, using these unifying topological descriptions, hierarchising their distributed computability (Section 2). In particular, this will extend the ACT to a much wider class of models than what has already been achieved in [9] and provide a means of comparing their distributed computability power without requiring ad hoc algorithmic reductions.

1. EXTENSIONS OF THE SPECTRAL SEMANTICS FOR DISTRIBUTED SYSTEMS

As described in Section 4 of the RS, the spectral semantics for round-based protocols established in [9] is based on considering a protocol as an endofunctor on the category $\mathbf{sComp}$ of simplicial complexes, along with a natural transformation describing the reachability constraints from initial configurations. Therein, we show that *round-based* protocols which are *colourless*, *full-information*, *wait-free* and *oblivious* can be interpreted in this way. In order to provide a global setting for distributed computability, we must extend our approach beyond its current scope, in order to describe communication models allowing more information and less scheduling constraints. The former concern leads to the first objective:

Objective 1. Extend spectral semantics to round-based communication models with identifiers and values, and characterise the models of round-based communication that fall within the scope of our functorial construction.

Identifiers and values. In [26], topological objects are used to characterise task-solvability for the IIS model in which processes have unique names, but as in the anonymous case, the result is extended to other models by means of algorithmic reductions. This is achieved using *chromatic simplicial complexes*, which can be identified with dimension-preserving simplicial maps $\mathcal{C} \rightarrow \mathcal{P}([n])$, where $[n]$ denotes the set with n elements. This corresponds to a categorical construction called the *slice category* $\mathbf{sComp}_n / \mathcal{P}([n])$, where $\mathbf{sComp}_n$ is the category of n -dimensional simplicial complexes and dimension-preserving maps. Projective limit constructions such as those employed to obtain the spectral semantics in the colourless case carry over to slice categories, providing a means of capturing chromatic models with the same methodology.

Alongside extending our results to the chromatic case, we should ensure that we can relate it to anonymous models. Given a chromatic model, the primitive *Give Input Value* (GIV) projects away the identifier in pairs (p, v) corresponding

to the local view v of a process p . This means that *multiplicity of values* is also projected out of the system. Therefore, in order to properly relate these two paradigms of distributed systems, a decoration of simplicial complexes with a countable set of values V should also be considered. This can also be achieved via slice categories, considering the category of spectral spaces over $\mathcal{P}(V)$ with the GIV primitive captured functorially. We expect that this will also allow us to consider *input-sensitive* protocols, *i.e.* those for which the action depends on the value, as opposed to the *oblivious* models thus far covered by our result.

Once a class of models is captured in the categorical setting, as in [9], we **obtain ACTs** for these classes by taking projective limits. Then, a further task in this direction is to **characterise the projective limits in these slice categories** in order to determine the correct definitions of *well-coloured spectral spaces* and *spectral spaces with values*. Among these, to **characterise the linear projective limits of (chromatic) simplicial complexes**, is an equally important step as it is important input for Objective 4, see below.

Functorial protocols. As indicated above, a number of interesting models of round-based distributed communication can be modeled with our endofunctor approach. This begs the long term question of characterising the protocols that can be modeled in this way, which we may call *functorial protocols*. As a starting point, I propose describing *communication graphs* [1] as transformations of input simplices, which would result in a functorial protocol. Communication graphs are directed graphs decorated with process names which encode a possible communication scheme. Languages of communication graphs capture a large class of round-based *message adversaries*, *i.e.* collections of allowed executions, without crash failures. Each graph in a given word corresponds to a round of communication in an iterated protocol. This is an initial step in **interpreting general message adversaries as endofunctors**. Further one may augment this approach by integrating *failure detectors* [31], used to model crashes in such message adversaries. This would not only allow us to **integrate crash-failures into this approach**, but is also relevant to Objective 4 below, in which the relationship between crash-failures and subspaces is considered.

The functorial approach of Objective 1 works well for describing round-based protocols, but fails for those exhibiting *asynchrony*. In order to capture the epistemic dynamic of any scheduling constraint, we want to associate a spectral space encoding epistemic information to general models of concurrent systems such as event structures, Petri nets, and Pratt's higher dimensional automata. My background in concurrency theory alongside my more recent experience with distributed systems puts me in a unique position to address this schism in the current approach. Relating the temporal aspects of concurrency to the epistemic dynamics of distributed systems constitutes a second objective in my research project:

Objective 2. Develop a formal correspondence between scheduling objects and spectral spaces, thereby extending our spectral semantics for distributed computing to asynchronous models of communication.

Spectral semantics of event structures. The approach given in [22] linking directed geometric semantics of concurrent systems to protocol complexes was a first indication that a connection to distributed computing could be made. Inspired by this work, Godard, Chalopin and I studied *interval orders*, partial orders describing executions of concurrent programs. Indeed, these were used as a bridge between the concurrent and distributed paradigms. These partial orders are also related to Winskel's *event structures* [32], which provide a general operational description of concurrent systems, as they can be thought of as glueings of interval orders along common prefixes. While studying the projective limit system we used in [9], I realised that I was in fact looking at an event structure. Indeed, the maps in the projective limit system define an ordering on local views which results in an event structure.

This raises the question of the inverse construction, namely, given an event structure, can we canonically construct a spectral space? Answering this question is the first step towards Objective 2. By construction, maximal antichains in event structures carry the structure of simplicial complexes. An initial task is to use this structure to associate a (non-linear) projective limit system of simplicial complexes to an event structure. This will allow us to **associate a spectral space to an event structure** which we then can use to **establish an ACT for event structures**. As in Objective 1, once such a construction is in place, we want to **characterise the pertinent classes of structures for this correspondence**. In particular, those spectral spaces arising as *non-linear* projective limits of simplicial complexes and, on the other side, the event structures that describe distributed systems. I also propose to **show that our spectral semantics generalise other recent topological approaches**. For example, in [4] the authors employ spaces defined via *ultra-metrics*, which are known to be examples of *Priestley topologies*. These can in turn be derived from the spectral topologies we employ [19, p. 199], allowing us to consolidate these two approaches.

Scheduling and epistemics. While event structures provide a general approach to concurrent systems, there has been a concerted effort over the past forty years to relate them to other concurrent objects such as Petri nets and Pratt’s higher dimensional automata (HDA) [28, 30, 2]. Part of my research project consists in using presheaf representations of such structures in order to build a correspondence theory between them. This would allow us to extend the pipeline proposed above, taking concurrent objects to epistemic spaces.

As opposed to automata, the languages of HDA consist not only of totally ordered strings of letters, but also of interval orders, mentioned above. Further, in [33], Winskel shows that event structures correspond to certain presheaves over the category of partial orders. I propose to **use categorical constructions to relate event structures to HDA**. It is my intuition that an event structure can be canonically defined from a HDA by glueing its language along the pasting scheme defined by the HDA. The converse construction works similarly, glueing the HDAs corresponding to each interval order in the event structure along their common prefixes. Importantly, this would provide a way of meaningfully **defining a notion of task or specification for such concurrency-based structures**. This is another important step in relating concurrent objects to distributed systems.

Connecting back my my earlier work, the relation between event structures and HDA can provide further insights on lattice quotients of multinomial lattices which I studied with Santocanale [13]. As mentioned in the RS, these quotients can be interpreted as homotopy classes of directed paths in an HDA. These homotopy classes correspond precisely to the languages of the HDA [18] so their interpretation as event structures, and thereby as diagrams of simplicial sets, then provide an alternative approach to the study of these lattices. Furthermore, in [30], Pratt relates HDA to Chu spaces, the latter providing a finer grained analysis of concurrent systems, in which quantales can be used to describe more complicated temporal orders than before, during and after. This relates to the convolution quantales we studied in [12].

So far, we have discussed *deterministic* distributed protocols, *i.e.* those in which each process is a deterministic finite automaton. Their randomised counterparts are equally of great interest. A notable result is that the consensus task is solvable in this paradigm, as proved in Ben-Or’s groundbreaking paper [6]. Moreover, the quantum paradigm is of interest to the distributed computing community, notably in establishing computability results in quantum models, with the particular aim of comparing their task-solvability to deterministic and randomised models. In recent literature, event structures are equipped with randomised or quantum behaviours [34, 27, 14], which, along with the results discussed above, provide a clear way of extending spectral semantics to these models. In order to obtain a fully universal approach to communication models, we should also include *Byzantine* communication models, in which processes may exhibit incorrect or malicious behaviours, but this is a longer-term goal.

Objective 3. Develop a topological approach to randomised and quantum models of distributed systems using the links between event structures and spectral spaces established in Objective 2, with an aim to eventually capture Byzantine models in a similar way.

Randomised and quantum distributed computing. The asymptotic nature of the spectral semantics seems highly pertinent to randomised protocols. Indeed, randomised solutions are determined by *eventually* producing a decision with high probability. Thus, a first step is to **equip the associated spectral spaces with a probability distributions** derived from the finite-time operations. This approach could be further investigated, and also extended to the quantum paradigm, by using recent literature in which event structures are equipped with randomised or quantum behaviours [34, 27, 14]. As above in the case of asynchronous models, another task under this objective is to use results and intuitions gained in the pursuit of Objective 2 in order to **translate these augmented event structures into diagrams of simplicial complexes** with extra structure and **study the projective limits we obtain**.

Byzantine communication. In the longer term, I would like to work with experts in event structures, such as Winskel and Clairambault, to **define a notion of Byzantine event structure**, for example by considering event structures with infinite branchings at each action of a Byzantine agent. This would lead to much more complex limit systems, in particular in the ordering of the associated spectral space. Then, using the relation between spectral semantics and event structures discussed above, we would be equipped to define tasks in this context using the insights into epistemics via spectral semantics gained in the case of regular event structures under Objective 2. This would then allow us to **characterise distributed computability for Byzantine, asynchronous models**.

2. COMPARING DISTRIBUTED MODELS VIA SPECTRAL SEMANTICS

As recalled in the introduction, comparing task-solvability for models of communication is currently limited to exhibiting *algorithmic simulations* between them. The existence of a simulation from model M to model M' means that any observable behaviour exhibited by M can be reproduced by M' . A bisimulation, *i.e.* a simulation in both directions, implies equivalent task-solvability. Using spectral semantics, we can provide *a priori* results implying the existence of a (bi)simulation, which would in addition provide intuitions for their algorithmic implementation.

Objective 4. Use topological results to establish the existence of (bi)simulations between models, identify sub-models, and relate this approach to categorical and logical methods of (bi)simulation.

(Bi)simulation via topological methods. Our generalised ACT results relate the solvability of a task $(\mathcal{I}, \mathcal{O}, \Delta)$ by a protocol (Π, π) to the existence of a spectral map $\delta : \Pi^\infty(\mathcal{I}) \rightarrow \mathcal{O}$. Thus, given another protocol (P, p) , if there exists a spectral map $\sigma : P^\infty(\mathcal{I}) \rightarrow \Pi^\infty(\mathcal{I})$, then any task solvable by Π is also solvable by P , since $\sigma \circ \delta$ is a spectral map. If σ is natural in $\mathcal{I}$, this is true for any input complex. Thus, a natural transformation from Π^∞ to P^∞ implies the existence of a simulation from Π to P . In particular, when σ is a *retract*, it establishes a bisimulation. Such a natural transformation can be interpreted as a mapping of executions of the protocol Π to those of P , thereby providing insight into how to implement them in practice.

I therefore propose to **determine under which hypotheses such a natural transformation exists and how it can be used to produce an algorithmic reduction**. For example, does the existence of a natural transformation from Π to P imply the existence of one from Π^∞ to P^∞ ? And how can we deduce an operational simulation from such a transformation? Answers to these questions would greatly simplify establishing simulations between communication models without relying on ad-hoc algorithmic constructions. Furthermore, the relationship between retracts and bisimulations, and the strong role the former play in algebraic topology, suggest the use of algebraic invariants to classify task-solvability.

Several more concrete tasks in this direction follow from the framework set up in Section 1. Indeed, a functorial GIV primitive would **provide a means of comparing the task-solvability of chromatic and colourless models**. Another such comparison, between asynchronous and round-based models, can be obtained by considering the maximal antichains of an event structure deduced from a linear projective limit system, as described in the context of Objective 2. Indeed, the event structures obtained from iterated models have more maximal antichains than those selected by the round-based grading. Using the topological simulation techniques discussed above, this would in particular allow us to **show that asynchronous and round-based protocols have equivalent task-solvability** for a large class of models. Finally, our unified topological approach would also allow us to extend results such as Ben-Or's randomised consensus, *i.e.* **determine precisely which advantages are gained by using randomised, as well as quantum models**.

Submodels as subspaces. Given a model of communication, it is of interest to consider a subset of the possible communication scenarios it permits. These submodels are, for example, useful in the study of *t-resilient models*, in which, contrary to the wait-free models described above, at most $n - t$ processes may crash. The only way to detect this condition is asymptotically, that is, a process must wait forever without hearing from another in order to conclude that it has crashed. In this sense, subsystems involving crashes can only be detected at the limit. It is therefore of interest to **characterise submodels in terms of subspaces** in the spectral semantics. Such a result will further extend spectral semantics to a yet wider class of models. In addition, given a model unable to solve some task, such a characterisation would allow us to identify maximal submodels able to solve the considered task. A starting point is offered by [15], in which Coutouly and Godard describe submodels of IIS via real Hausdorff topologies, considering a quotient of the space of executions. The latter is not even T_0 , so our spectral approach, which lies between these two approach in terms of separability, seems suited to describe such submodels.

Epistemic logics and open maps. In the 1990s, epistemic modal logic [17] was widely used to determine properties of distributed systems [25]. Since then, the topological approach spearheaded by Herlihy and Shavit has proved more effective, especially in characterising task-solvability. In a recent article [20], the authors relate these two approaches, showing that by interpreting protocols as endofunctors on $\mathbf{sComp}$, as was done by myself and Godard in [9], naturally gives rise to algebras, which are then shown to correspond to the semantics of a temporal-epistemic logic. As mentioned in the RS, we employed Stone duality to obtain our computability result. The logical approach described above is highly related, and a short-term task is to **show that the algebras of [20] are precisely the dual lattices to the spectral semantics from [9]**. This will consolidate the temporal logic approach to multi-agent systems and the more modern topological approach.

From there, I propose to **construct bisimulations via logical methods** using recent works relating co-algebras to bisimulations [7]. Indeed, this approach can be applied to the algebras of [20], providing a means of translating between

the topological bisimulations described above and those obtained from the (co)algebraic approach. *Open maps* [3] have also been used to prove bisimulation in the context of event structures [33] and HDA [16]. Relating these *temporal bisimulations* to those described above would further my goal of connecting the scheduling and epistemic aspects of distributed protocols. I therefore propose to **determine if and when such temporal bisimulations imply epistemic bisimulations**. A starting point would be [35], in which the authors develop logical bisimulations for HDA. Finally, I propose to **compare these bisimulation techniques to operational approaches**, for example in the recent article [5] in which the *Heard-of* and the *Asynchronous Message Passing* models are shown to have equivalent task-solvability.

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